UMC

2303.TW | Taiwan Stock Exchange | 2026-10-07

Q4 2027 model ceiling: TWD 132.63

Analytical Base 40/60 = 67 | Specialty foundry recovery with photonics and packaging options

20x rolling Q+next3 TIFRS operating profit, divided by window-end Taiwan common shares including conservative new-convertible dilution. No investment-gain or excess-cash bonus.

Coverage initiated at the owner's request. No holding was established by this research. Existing portfolio Base overrides remain protected.

All operating values are in TWD m and shares in millions unless stated otherwise. Forecasts are analytical assumptions; issuer guidance and actual results are identified separately.

Decision in the model

Active horizon	Earnings window	F4Q OP	Multiple	Diluted shares	Ceiling
Q4 2027	Q4 2027 through Q3 2028	85,633.9	20x	12,913.422	132.63

The ceiling is a model valuation, not a guaranteed trading price or a literal net-income P/E. Operating branches are weighted once. Where debt, preferred claims, conversion or equity floors make value nonlinear, the model values each coherent operating and funding state first, then weights those common-share outcomes.

Economic scope and matching claims

TIFRS OP includes depreciation, recurring R&D, SG&A and SBC; other operating income is explicitly modeled. Interest, tax and investment gains are outside OP. TIFRS is Taiwan-endorsed IFRS, not a claim of full IASB IFRS equivalence.. 20x rolling Q+next3 TIFRS operating profit, divided by window-end Taiwan common shares including conservative new-convertible dilution. No investment-gain or excess-cash bonus.

Financial claims and diluted shares follow the company-specific schedule below. Debt, preferred and minority claims are counted once. All financial model tables use TWD m unless a different unit is stated.

Thesis

Research date: October 7, 2026. Taiwan common stock 2303; financials and model values in NT$ million unless stated otherwise.

The Q4 2027 model ceiling is **NT$132.63 per Taiwan common share**, using **20x forward-four-quarter TIFRS operating profit**. That quarter includes Q4 2027 through Q3 2028. The analytical Base is **67/100**, from 40/60 framework points. This is a modeled ceiling, not a forecast of the market price or a promised return. New-platform economics and post-2028 results have lower confidence than the near-term foundry recovery.

UMC can earn more from established specialty wafers before photonics becomes large. The investment case rests on accepted wafer output, economically useful process differentiation and repricing that survives higher costs. Silicon photonics and advanced packaging add credible opportunities, but tools, qualification, depreciation and customer economics govern the speed of profit recognition.

Financial starting point

The last complete quarter is Q2 2026. Revenue was NT$68,733 m, gross profit NT$22,323 m and operating profit NT$14,950 m. The operating reconciliation is gross profit less NT$7,906 m expenses plus NT$533 m other operating income. Parent net income of NT$42,260 m benefited heavily from nonoperating investment income; it is not the earnings base for this valuation. Q2 shipments were 1.129 m 12-inch-equivalent wafers and utilization was 85%. Original Q2 report

July, August and September released sales total **NT$74,239.117 m**. The first model row therefore contains actual quarterly revenue but estimated margins, operating expenses and cash flow. It is not a released Q3 earnings result. Management's July guide was high-single-digit shipment growth, firm dollar ASP, mid-30% gross margin and utilization above 90%. Monthly sales, Q2 results presentation

The reference Q3 allocation produces roughly 35% gross margin. It is calibrated to the released revenue and management's range, not reverse-engineered from an EPS forecast. For context, 2025 revenue was NT$237.553 bn and operating profit NT$43.949 bn. The rapid 2026 recovery starts from that lower utilization base.

Operating Results Summary

Unit: NT$ M

	2026 2Q / Jun.	2026 1Q / Mar.	2025 2Q / Jun.
Operating Revenues	68,733	61,038	58,758
Net Income Attributable to Shareholders of the Parent	42,260	16,171	8,903
EPS (NTD)	3.39	1.29	0.71
Cash and Cash Equivalents	124,706	109,019	111,994
Total Assets	665,973	599,932	548,144
Total Liabilities	222,049	193,196	211,098
Total Equity	443,924	406,736	337,046
Foundry Segment:			
Wafer Shipments (in thousands-12" wafer eq.)	1,129	1,021	967
Utilization (%)	85%	79%	76%

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Original issuer exhibit | Q2 2026 original operating summary; shipment units are thousands of 12-inch equivalents.

Statements of Comprehensive Income - QoQ

Unit: NT$ M

	2Q26	%	1Q26	%	Change %
Operating Revenues	68,733	100.0	61,038	100.0	12.6
Gross Profit	22,323	32.5	17,818	29.2	25.3
Operating Expenses	(7,906)	(11.5)	(7,099)	(11.6)	11.4
Net Other Operating Income and Expenses	533	0.8	557	0.9	(4.3)
Operating Income	14,950	21.8	11,276	18.5	32.6
Net Non-operating Income and Expenses	30,236	44.0	5,367	8.8	463.3
Income Tax Expense	(2,962)	(4.3)	(527)	(0.9)	462.3
Net Income	42,224	61.4	16,117	26.4	162.0
Net Income Attributable to Shareholders of the Parent	42,260	61.5	16,171	26.5	161.3
EPS (NTD)	3.39		1.29		
EPADS (USD)	0.537		0.204		

Notes:
1. 12,475 million and 12,491 million shares were used in EPS calculations for 2Q26 and 1Q26, respectively.
2. Exchange rates (USD/NTD) for 2Q26 and 1Q26 were 31.59 and 31.63, respectively.
3. Sums may not equal totals due to rounding.

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Original issuer exhibit | Q2 2026 operating profit is NT$14.950bn; investment gains are outside operating profit.

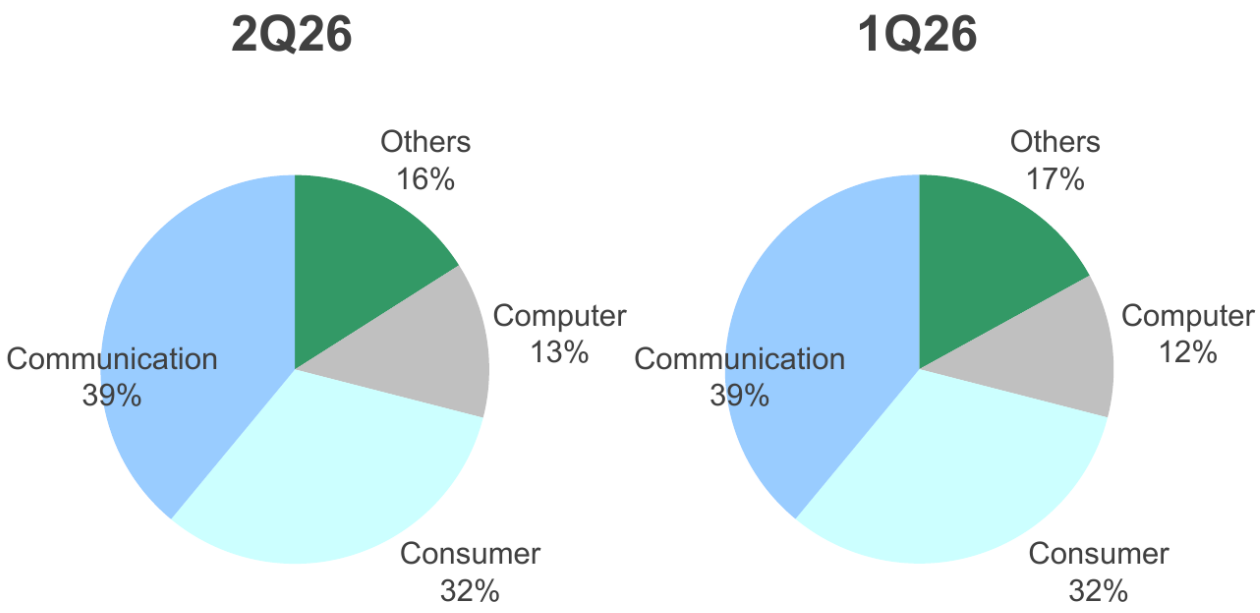
What UMC sells

The six modeled revenue vectors are mutually exclusive: mainstream 22/28 nm;40/65 nm power, RF and connectivity;90 nm-and-older; silicon-photonics/TFLN wafer processing; advanced-packaging/DTC/bridge services; and recognized 12/14 nm service revenue. Photonics and packaging are carved out of node-based sales, rather than added on top of the entire published node mix. Application categories such as communication and consumer are cross-cuts of the same sales.

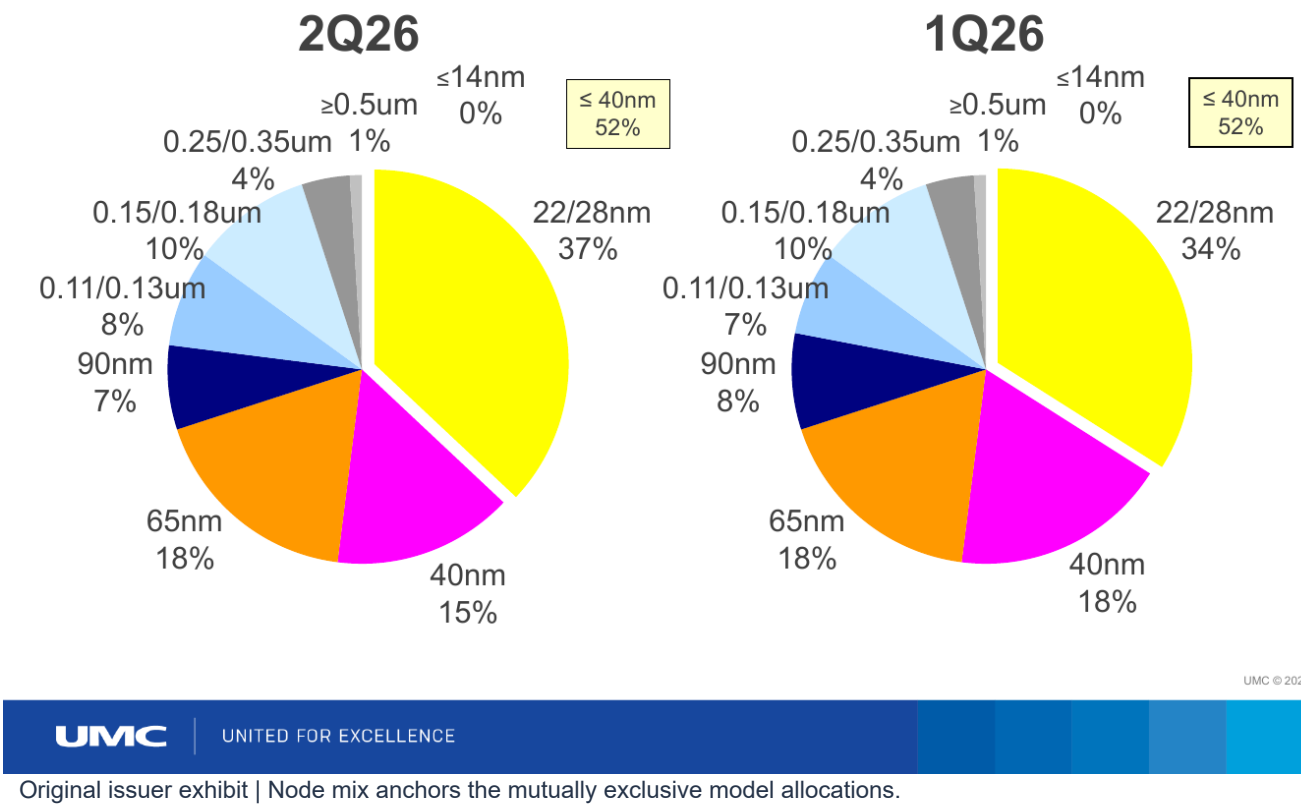
The mainstream business is valuable because an analogue or high-voltage device cannot always migrate economically to the smallest transistor. Device voltage, leakage, RF performance, reliability, qualification history and mask/design cost matter. Customers can change foundries, but a qualified process and library are not interchangeable with an available square metre of cleanroom.

The reference legacy wafer sales imply about 5.10 m,5.25 m,5.32 m and 5.28 m annual 12-inch-equivalent shipments in 2027-30 at modeled blended prices of NT$62 k,64 k,66 k and 68 k. Installed-capacity assumptions are 5.50 m,5.85 m,6.10 m and 6.30 m equivalents annually. These are cross-checks, not disclosed capacity guidance. Photonic and packaging wafer-equivalent requirements must fit the residual capacity or dedicated additions; they are not costless extra throughput. Rising mix can lift the average price even when a customer's like-for-like wafer price is unchanged.

Foundry Segment Sales Breakdown by Application



Original issuer exhibit | Application mix is a cross-cut of node sales, not an additional revenue pool.



Original issuer exhibit | Node mix anchors the mutually exclusive model allocations.

Capacity, qualification and the cost of waiting

UMC's July table shows Q3 installed capacity of 1.325 m 12-inch equivalents, with Singapore 12 i increasing to 192 k from 172 k. This is an installed-production schedule, unlike an approved future building. Customer acceptance, available tools, cycle time and good-die yield determine the sellable result.

A long wait for qualified power or connectivity silicon can justify a higher wafer price: the customer's lost system revenue may be much larger than the incremental component cost. But existing fixed-price agreements, alternatives and cost inflation limit captured rent. The active price branch uses 70% probability of a 4% premium on 50% of 2027 legacy sales, with explicit cost retention and later fade. It does not assume a universal 8-10% price hike. In April management confirmed price-adjustment letters but did not validate the percentage suggested by an analyst. April management discussion

The NT$148.680 bn board-authorized investment budget is phased capital spending, not revenue or immediately usable capacity. The separate 2026 cash-capex guide is US$2 bn. Adding both as simultaneous cash outlays would double-count the approved program. July board filing

Quarterly Capacity

Unit: in thousands

FAB		4Q25	1Q26	2Q26	3Q26E
WTK	(6")	80	78	80	80
8A	(8")	215	212	215	215
8C	(8")	125	123	125	125
8D	(8")	118	116	118	118
8E	(8")	131	129	131	131
8F	(8")	146	144	146	146
8S	(8")	117	115	117	117
8N	(8")	250	246	250	250
12A	(12")	409	402	409	409
12i	(12")	172	169	172	192
12X	(12")	95	93	95	95
12M	(12")	119	117	119	119
Total	(12" eq.)	1,305	1,283	1,305	1,325

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Original issuer exhibit | Installed wafer capacity differs from accepted shipments and future cleanroom capacity.

New platforms and competition

July's SILITH announcement established the first mass-production 12-inch PIC wafer delivery. Its 200 G/lane platform and general-customer 2027 availability are distinct milestones. SILITH's cumulative 8 m 100 G/200 G PIC shipments are company-level history, not 8 m UMC wafers. The customer and foundry earn different portions of a transceiver's revenue. UMC/SILITH release

At the modeled NT$200 k recognized revenue per accepted PIC/TFLN wafer equivalent, active volumes are 26 k,44.5 k,70.5 k and 93 k in 2027-30. ASP is an explicit NT$150 k-250 k sensitivity, not an issuer quote. The branch pays for processing, ordinary rejects, depreciation, application engineering and qualification. It does not infer revenue from a full module price or from the area ratio between 12-inch and 8-inch wafers.

The HyperLight/Wavetek/Jabil relationship connects material/process technology to module integration. TFLN can help high-speed modulation, but requires its own bonding, optical coupling, handling and yield discipline. A lower-loss modulator can reduce laser-power requirements in particular designs; it does not prove every architecture eliminates high-power external lasers. Original collaboration

Competitors include established specialty-photonics foundries such as Tower and GlobalFoundries, broader integrated offerings from TSMC, and alternative electro-optic materials and architectures. Qualification and measured system performance determine substitution. A large wafer diameter alone is not proof of the lowest delivered cost per functioning optical channel.

The annual roadmap documents power, RF, embedded-memory and packaging work. These are useful adjacent capabilities, but the model books paid process services only once. No full CoWoS platform, complete AI accelerator, substrate wallet or third-party module revenue is assigned to UMC. 2025 annual report

July management remarks put 12 nm customer tape-outs/pilot in 2027 and meaningful production in 2028, later than a broad reading of the original collaboration announcement. They also described construction lead times beyond 20 months, depreciation growth and more constructive 2027 pricing. These statements control our timing; an approved shell is not a production award. Management's approximate US$300 m 2026 AI exposure and greater-than US$1 bn aspiration in three years are cross-checks across power, connectivity and packaging, not extra revenue vectors. July management discussion

Intel's contract revenue/profit split is not public. Our small service-revenue proxy starts with pilot activity and grows in 2028; it must be replaced when recognized revenue and the economic share are known. No leading-edge 3 nm deal, fab purchase or additional AI TAM is priced into the central case. Undefined deal perimeter and economics prevent a meaningful profit bridge; that is different from dismissing a credible event because it lacks confirmation.



Original issuer exhibit | Original annual report page 5 technology excerpt; July management timing supersedes the earlier 12nm volume-production expectation.

Probability-weighted operating bridge

The reference already includes improving utilization, normal 22 nm mix, existing packaging production and a gradual photonics ramp. Active deltas are residuals after that overlap adjustment. Probabilities are our judgments, not management forecasts. The Q4 2027 forward window is:

Increment above reference	Probability	Weighted revenue	Weighted OP
VALUE-PRICING	70%	3,765.4	2,259.3
PIC-QUALIFIED-VOLUME	60%	2,082.0	624.6
PACKAGING-ACCEPTANCE	45%	180.0	41.4
INTEL-12 NM	50%	582.8	87.4
MATURE-PRICE-DOWNSIDE	30%	-797.8	-678.1

The reference ceiling is NT$129.01; the fully costed active branches add NT$3.62, reaching NT$132.63. No analyst price target or blanket AI multiple is imported. The downside price branch addresses residual mature-node competition; it does not also subtract a second demand or utilization shock.

Production depreciation is embedded in gross margins. Total modeled annual depreciation rises from NT$65.0 bn in 2026 to 72.8 bn,81.536 bn,87.0 bn and 91.0 bn. R&D is 7.5% of reference revenue; selling/administration moves from 4.2% toward 3.8%. A separate declining other-operating-income proxy reconciles the statutory definition. Incremental branches fund additional application support; they do not receive corporate R&D for free.

The later path is deliberately a bounded operating scenario. A 2028 margin upgrade needs evidence that achieved prices and mix more than cover new depreciation and energy costs. Positive EBITDA commentary alone does not establish that result.

Funding and dilution

June cash was NT$124.706 bn, ordinary borrowings/bonds NT$57.533 bn and the declared dividend payable NT$32.704 bn. Noncurrent lease liabilities were NT$5.277 bn; current lease detail and the full maturity/deposit notes remain gaps. Investment assets are excluded from the financing solution and valuation bonus. Original balance and cash-flow statements

The August capital-reduction filing gives 12,539.542899 m outstanding common shares after canceled restricted shares. This differs from weighted-average EPS shares. The model adds a 0.15% annual award reserve and conservative full conversion of the new bonds. One NYSE ADS represents five Taiwan common shares; this report's ceilings are in TWD per common share. Share cancellation and domestic conversion terms

September's domestic bond has NT$4 bn face value at NT$130.7 conversion price: 30.604438 m potential shares. NT$4.792799 bn was cash raised, not the face-value conversion numerator. The exchange announcement identifies code 23032 and September 3 listing. Exchange notice, Issuer proceeds filing

The October 5 terms schedule a US$1.8 bn overseas issue for October 13, split equally into two zero-coupon tranches. Fixed conversion FX is 31.753, with prices 202.06 and 179.19; those create 300.914425 m potential common shares. The amended text still labels the second price A; treating it as B is a flagged inference from the two-tranche structure. A has an October 2027 put; B matures April 2028. Proceeds are not yet settled as of this research date. Amended original filing

The base uses a fully-if-converted denominator, while the funding stress tests cash repayment as an alternative; both are not charged to the same outcome. Q4 2027's matched end-window denominator is **12,913.422400 m** shares. No forecast buyback or undisclosed equity rescue is assumed.

Even if all new bonds are cash-redeemed at the modeled early dates under base operations, minimum cash is NT$87.885bn. Full conversion is therefore a conservative share-reserve convention, not a claim that conversion below the strike is economically likely.

The cash roll includes taxes, working-capital growth, dividends, leases, ordinary debt payments and capex. Base minimum cash is NT$92.467 bn. Without the October issue, it is NT$86.367 bn, so the central investment program is not dependent on those unreceived proceeds. A severe case cuts operating profit 35%, raises capex 25%, increases working-capital needs and cash-redeems the CBs at their earliest modeled dates; the resulting funding gap is NT$68.973 bn. A no-new-issue case with immediate repayment of all June ordinary debt reaches NT$73.522 bn. Those gaps require spending restraint, dividend reduction, debt or equity decisions; none is silently assumed solved.

2026 Capital Expenditure Plan

8"	12"	Total
10%	90%	2.0 billion USD

Original issuer exhibit | 2026 cash capital expenditure guidance increased to US$2bn.

Base score, catalysts and failure tests

Core 15/20, cycle 7/10, TAM 7/10, catalysts 4/5, macro 7/10, alpha 4/5 and risks-4 total 40/60, or 67/100. The score recognizes a qualified specialty business and improving earnings while reserving for capital intensity, China competition and unverified new-platform economics. It does not depend on a rising share price.

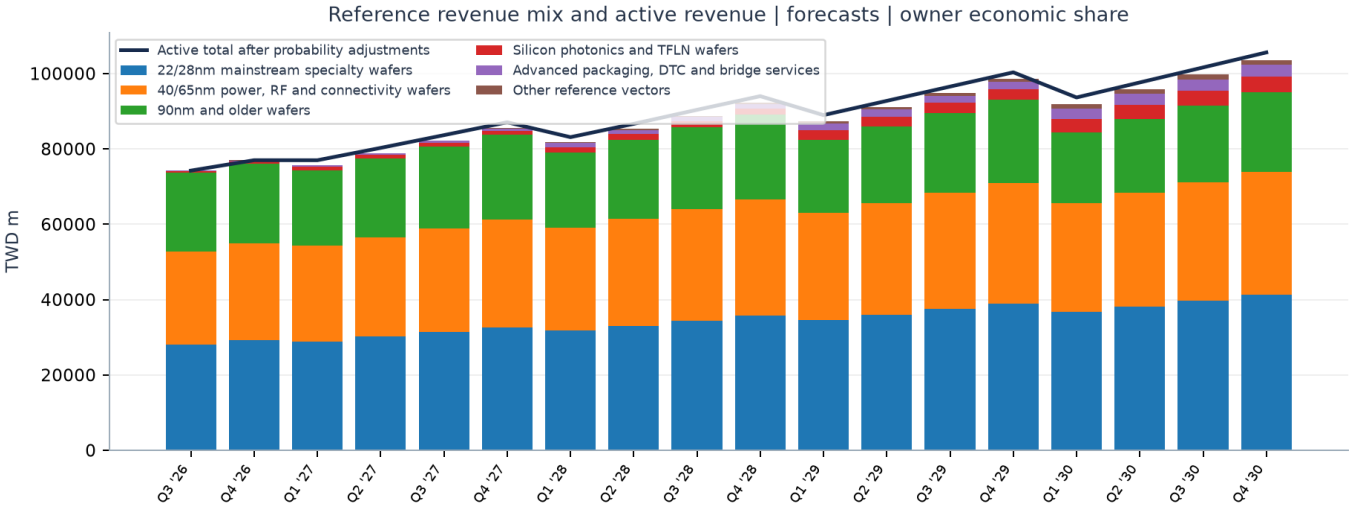
The next quarterly disclosure should reconcile achieved wafer prices, gross margin and spending with the monthly revenue strength. Through 2027, monitor accepted photonics shipments, general-platform customers and 12 nm tape-outs. Through 2028, focus on installed tools, usable output, depreciation and paid packaging deliveries. Escalate only the affected driver when evidence changes.

The thesis weakens if price rises merely repay input inflation, non-AI inventory reverses, qualified capacity fails to earn its capital cost, or emerging platforms remain technically attractive without material recognized profit. Revisit the model if photon/wafer units or partner recognition differ from the stated proxies, if capex exceeds the funded case, or if bond settlement/conversion terms change. Keep the lower-confidence 2030 path separate from near-term actuals.

Score arithmetic

Category	Points	Evidence / treatment
Core thesis	15	Qualified specialty foundry franchise and existing cash generation; new photonics customer proof remains small.
Cycle	7	Utilization recovery and pricing are positive, while depreciation and non-AI inventory are offsets.
TAM	7	Photonics, power and packaging broaden paid wafer/process services; no end-market TAM shortcut.
Catalysts	4	October29 quarterly results, achieved prices, 2027 general PIC platform and12 nm tape-outs,2028 capacity acceptance.
Macro	7	AI connectivity/power growth and geographic diversification are offset by China capacity, Taiwan risk and FX.
Alpha	4	Separate observed sales, mix, new-price rent, photonic die versus wafer counts, gross versus net partner economics, and coupon versus dilution cost.
Risk classes	-4	Qualification and installation delays, spending before revenue, contract economics, price competition, customer inventory and capital-market terms.
Total	40/60	Rounded Base 67. Both artifact and new workbook row use this score; this is not a manual override.

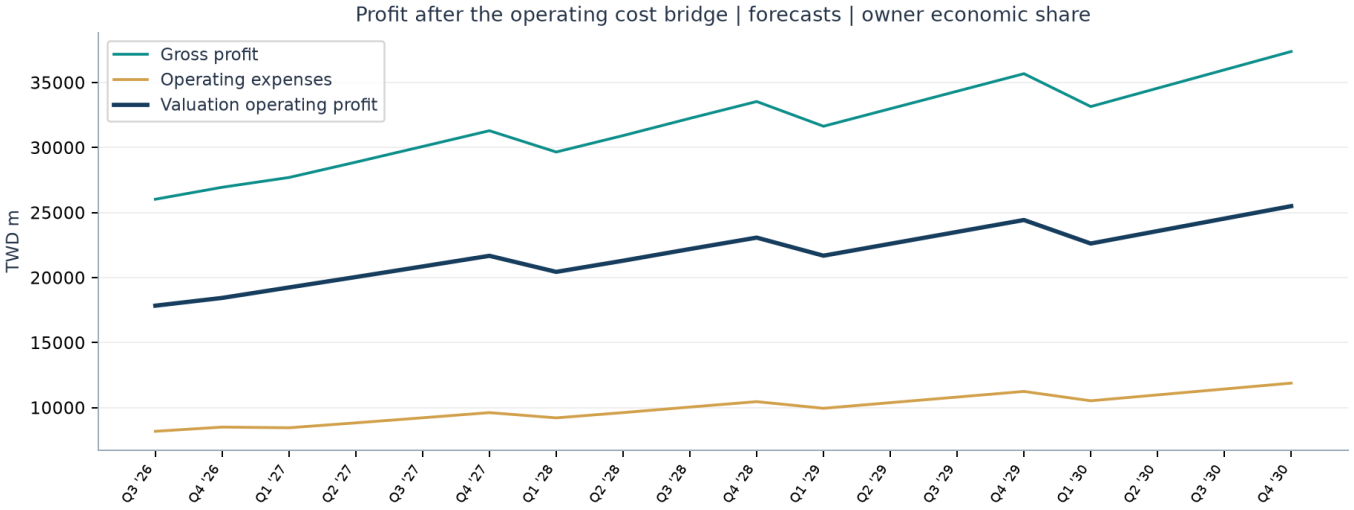
Revenue and business mix



Reference business vectors sum to the reference; the active total adds only probability-weighted net changes. Product allocations may be analytical proxies where issuers do not report product margins.

Quarter	Reference revenue	Weighted net change	Active revenue
Q3 2026	74,239.1	0.0	74,239.1
Q4 2026	77,000.0	0.0	77,000.0
Q1 2027	75,670.0	1,321.6	76,991.6
Q2 2027	78,890.0	1,377.9	80,267.9
Q3 2027	82,170.0	1,479.1	83,649.1
Q4 2027	85,470.0	1,595.4	87,065.4
Q1 2028	81,827.0	1,290.8	83,117.8
Q2 2028	85,309.0	1,345.7	86,654.7
Q3 2028	88,791.0	1,580.6	90,371.6
Q4 2028	92,273.0	1,725.5	93,998.5
Q1 2029	87,420.0	1,548.2	88,968.2
Q2 2029	91,140.0	1,614.1	92,754.1
Q3 2029	94,860.0	1,680.0	96,540.0
Q4 2029	98,580.0	1,745.9	100,325.9
Q1 2030	91,885.0	1,767.9	93,652.9
Q2 2030	95,795.0	1,843.1	97,638.1
Q3 2030	99,705.0	1,918.4	101,623.4
Q4 2030	103,615.0	1,993.6	105,608.6

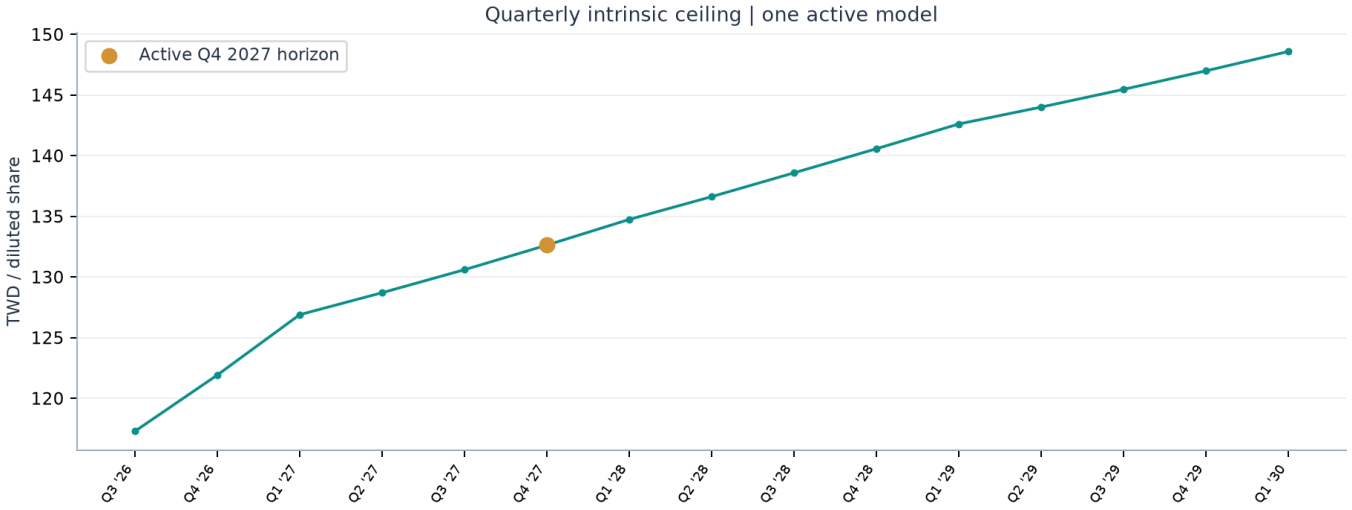
Gross profit, operating expenses and earnings



Operating profit equals gross profit less operating expenses and any separately identified attribution reserve. Figures are forecasts, not reported historical quarterly outcomes.

Quarter	Revenue	GP	Opex	Valuation OP	OP margin
Q3 2026	74,239.1	26,032.2	8,186.0	17,846.3	24.0%
Q4 2026	77,000.0	26,953.0	8,509.0	18,444.0	24.0%
Q1 2027	76,991.6	27,705.5	8,457.9	19,247.6	25.0%
Q2 2027	80,267.9	28,884.4	8,834.8	20,049.6	25.0%
Q3 2027	83,649.1	30,088.9	9,225.5	20,863.4	24.9%
Q4 2027	87,065.4	31,301.8	9,620.7	21,681.1	24.9%
Q1 2028	83,117.8	29,664.6	9,218.1	20,446.5	24.6%
Q2 2028	86,654.7	30,926.9	9,623.1	21,303.8	24.6%
Q3 2028	90,371.6	32,252.2	10,049.8	22,202.5	24.6%
Q4 2028	93,998.5	33,546.1	10,465.6	23,080.5	24.6%
Q1 2029	88,968.2	31,645.9	9,954.8	21,691.1	24.4%
Q2 2029	92,754.1	32,992.5	10,386.9	22,605.6	24.4%
Q3 2029	96,540.0	34,339.1	10,819.0	23,520.1	24.4%
Q4 2029	100,325.9	35,685.8	11,251.2	24,434.6	24.4%
Q1 2030	93,652.9	33,159.3	10,533.0	22,626.3	24.2%
Q2 2030	97,638.1	34,570.3	10,985.4	23,584.9	24.2%
Q3 2030	101,623.4	35,981.3	11,437.9	24,543.4	24.2%
Q4 2030	105,608.6	37,392.3	11,890.4	25,502.0	24.1%

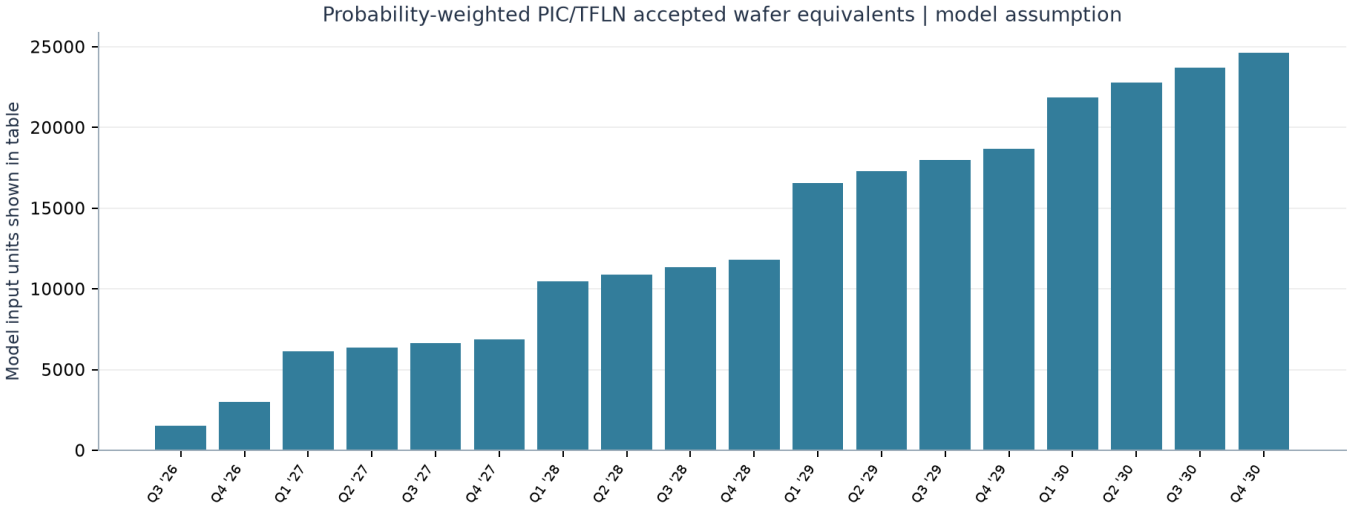
The complete ceiling path



Quarter Q prices Q plus the next three calendar quarters. All fifteen ceilings link to the same operating rows used in the workbook. Outer quarters have lower evidence visibility.

Quarter	Four-quarter earnings window	F4Q OP	Multiple	Shares	Ceiling
Q3 2026	Q3 2026 to Q2 2027	75,587.4	20x	12,889.871	117.28
Q4 2026	Q4 2026 to Q3 2027	78,604.6	20x	12,894.578	121.92
Q1 2027	Q1 2027 to Q4 2027	81,841.7	20x	12,899.286	126.89
Q2 2027	Q2 2027 to Q1 2028	83,040.6	20x	12,903.997	128.71
Q3 2027	Q3 2027 to Q2 2028	84,294.8	20x	12,908.709	130.60
Q4 2027	Q4 2027 to Q3 2028	85,633.9	20x	12,913.422	132.63
Q1 2028	Q1 2028 to Q4 2028	87,033.2	20x	12,918.138	134.75
Q2 2028	Q2 2028 to Q1 2029	88,277.8	20x	12,922.855	136.62
Q3 2028	Q3 2028 to Q2 2029	89,579.6	20x	12,927.574	138.59
Q4 2028	Q4 2028 to Q3 2029	90,897.3	20x	12,932.295	140.57
Q1 2029	Q1 2029 to Q4 2029	92,251.4	20x	12,937.018	142.62
Q2 2029	Q2 2029 to Q1 2030	93,186.6	20x	12,941.742	144.01
Q3 2029	Q3 2029 to Q2 2030	94,165.9	20x	12,946.468	145.47
Q4 2029	Q4 2029 to Q3 2030	95,189.2	20x	12,951.196	147.00
Q1 2030	Q1 2030 to Q4 2030	96,256.6	20x	12,955.926	148.59

The operating constraint behind the forecast



The driver series is a stated model assumption, not a disclosed commitment. It must be evaluated with the qualification, customer capture, cost and overlap discussion.

Quarter	Probability-weighted PIC/TFLN accepted wafer equivalents	Status
Q3 2026	1,500.000	Revenue actual from monthly releases; margins and costs estimated
Q4 2026	3,000.000	Forecast assumption
Q1 2027	6,110.000	Forecast assumption
Q2 2027	6,370.000	Forecast assumption
Q3 2027	6,630.000	Forecast assumption
Q4 2027	6,890.000	Forecast assumption
Q1 2028	10,457.500	Forecast assumption
Q2 2028	10,902.500	Forecast assumption
Q3 2028	11,347.500	Forecast assumption
Q4 2028	11,792.500	Forecast assumption
Q1 2029	16,567.500	Forecast assumption
Q2 2029	17,272.500	Forecast assumption
Q3 2029	17,977.500	Forecast assumption
Q4 2029	18,682.500	Forecast assumption
Q1 2030	21,855.000	Forecast assumption
Q2 2030	22,785.000	Forecast assumption
Q3 2030	23,715.000	Forecast assumption
Q4 2030	24,645.000	Forecast assumption

Quarterly revenue-vector derivation

All rows below are reference inputs. Gross profit = vector revenue x the stated gross margin. The next table applies the separately weighted evidence deltas. A product-level proxy is a model assumption even when the group total is calibrated to guidance.

Quarter	Revenue vector	Revenue	Gross margin	Gross profit
Q3 2026	22/28nm mainstream specialty wafers	28,100.0	40.0%	11,240.0
Q3 2026	40/65nm power, RF and connectivity wafers	24,700.0	34.0%	8,398.0
Q3 2026	90nm and older wafers	20,989.1	30.0%	6,296.7
Q3 2026	Silicon photonics and TFLN wafers	300.0	20.0%	60.0
Q3 2026	Advanced packaging, DTC and bridge services	150.0	25.0%	37.5
Q3 2026	12/14nm recognized service revenue	0.0	10.0%	0.0
Q4 2026	22/28nm mainstream specialty wafers	29,200.0	40.0%	11,680.0
Q4 2026	40/65nm power, RF and connectivity wafers	25,700.0	34.0%	8,738.0
Q4 2026	90nm and older wafers	21,100.0	30.0%	6,330.0
Q4 2026	Silicon photonics and TFLN wafers	600.0	20.0%	120.0
Q4 2026	Advanced packaging, DTC and bridge services	300.0	25.0%	75.0
Q4 2026	12/14nm recognized service revenue	100.0	10.0%	10.0
Q1 2027	22/28nm mainstream specialty wafers	28,905.0	40.0%	11,562.0
Q1 2027	40/65nm power, RF and connectivity wafers	25,380.0	35.0%	8,883.0
Q1 2027	90nm and older wafers	19,975.0	30.0%	5,992.5
Q1 2027	Silicon photonics and TFLN wafers	940.0	35.0%	329.0
Q1 2027	Advanced packaging, DTC and bridge services	470.0	32.0%	150.4
Q1 2027	12/14nm recognized service revenue	0.0	20.0%	0.0
Q2 2027	22/28nm mainstream specialty wafers	30,135.0	40.0%	12,054.0
Q2 2027	40/65nm power, RF and connectivity wafers	26,460.0	35.0%	9,261.0
Q2 2027	90nm and older wafers	20,825.0	30.0%	6,247.5
Q2 2027	Silicon photonics and TFLN wafers	980.0	35.0%	343.0
Q2 2027	Advanced packaging, DTC and bridge services	490.0	32.0%	156.8
Q2 2027	12/14nm recognized service revenue	0.0	20.0%	0.0

Quarter	Revenue vector	Revenue	Gross margin	Gross profit
Q3 2027	22/28nm mainstream specialty wafers	31,365.0	40.0%	12,546.0
Q3 2027	40/65nm power, RF and connectivity wafers	27,540.0	35.0%	9,639.0
Q3 2027	90nm and older wafers	21,675.0	30.0%	6,502.5
Q3 2027	Silicon photonics and TFLN wafers	1,020.0	35.0%	357.0
Q3 2027	Advanced packaging, DTC and bridge services	510.0	32.0%	163.2
Q3 2027	12/14nm recognized service revenue	60.0	20.0%	12.0
Q4 2027	22/28nm mainstream specialty wafers	32,595.0	40.0%	13,038.0
Q4 2027	40/65nm power, RF and connectivity wafers	28,620.0	35.0%	10,017.0
Q4 2027	90nm and older wafers	22,525.0	30.0%	6,757.5
Q4 2027	Silicon photonics and TFLN wafers	1,060.0	35.0%	371.0
Q4 2027	Advanced packaging, DTC and bridge services	530.0	32.0%	169.6
Q4 2027	12/14nm recognized service revenue	140.0	20.0%	28.0
Q1 2028	22/28nm mainstream specialty wafers	31,725.0	40.0%	12,690.0
Q1 2028	40/65nm power, RF and connectivity wafers	27,260.0	35.0%	9,541.0
Q1 2028	90nm and older wafers	19,975.0	29.0%	5,792.8
Q1 2028	Silicon photonics and TFLN wafers	1,527.5	40.0%	611.0
Q1 2028	Advanced packaging, DTC and bridge services	1,057.5	35.0%	370.1
Q1 2028	12/14nm recognized service revenue	282.0	28.0%	79.0
Q2 2028	22/28nm mainstream specialty wafers	33,075.0	40.0%	13,230.0
Q2 2028	40/65nm power, RF and connectivity wafers	28,420.0	35.0%	9,947.0
Q2 2028	90nm and older wafers	20,825.0	29.0%	6,039.2
Q2 2028	Silicon photonics and TFLN wafers	1,592.5	40.0%	637.0
Q2 2028	Advanced packaging, DTC and bridge services	1,102.5	35.0%	385.9
Q2 2028	12/14nm recognized service revenue	294.0	28.0%	82.3
Q3 2028	22/28nm mainstream specialty wafers	34,425.0	40.0%	13,770.0
Q3 2028	40/65nm power, RF and connectivity wafers	29,580.0	35.0%	10,353.0
Q3 2028	90nm and older wafers	21,675.0	29.0%	6,285.8

Quarter	Revenue vector	Revenue	Gross margin	Gross profit
Q3 2028	Silicon photonics and TFLN wafers	1,657.5	40.0%	663.0
Q3 2028	Advanced packaging, DTC and bridge services	1,147.5	35.0%	401.6
Q3 2028	12/14nm recognized service revenue	306.0	28.0%	85.7
Q4 2028	22/28nm mainstream specialty wafers	35,775.0	40.0%	14,310.0
Q4 2028	40/65nm power, RF and connectivity wafers	30,740.0	35.0%	10,759.0
Q4 2028	90nm and older wafers	22,525.0	29.0%	6,532.2
Q4 2028	Silicon photonics and TFLN wafers	1,722.5	40.0%	689.0
Q4 2028	Advanced packaging, DTC and bridge services	1,192.5	35.0%	417.4
Q4 2028	12/14nm recognized service revenue	318.0	28.0%	89.0
Q1 2029	22/28nm mainstream specialty wafers	34,545.0	40.5%	13,990.7
Q1 2029	40/65nm power, RF and connectivity wafers	28,435.0	34.5%	9,810.1
Q1 2029	90nm and older wafers	19,505.0	28.0%	5,461.4
Q1 2029	Silicon photonics and TFLN wafers	2,467.5	43.0%	1,061.0
Q1 2029	Advanced packaging, DTC and bridge services	1,762.5	38.0%	669.8
Q1 2029	12/14nm recognized service revenue	705.0	32.0%	225.6
Q2 2029	22/28nm mainstream specialty wafers	36,015.0	40.5%	14,586.1
Q2 2029	40/65nm power, RF and connectivity wafers	29,645.0	34.5%	10,227.5
Q2 2029	90nm and older wafers	20,335.0	28.0%	5,693.8
Q2 2029	Silicon photonics and TFLN wafers	2,572.5	43.0%	1,106.2
Q2 2029	Advanced packaging, DTC and bridge services	1,837.5	38.0%	698.2
Q2 2029	12/14nm recognized service revenue	735.0	32.0%	235.2
Q3 2029	22/28nm mainstream specialty wafers	37,485.0	40.5%	15,181.4
Q3 2029	40/65nm power, RF and connectivity wafers	30,855.0	34.5%	10,645.0
Q3 2029	90nm and older wafers	21,165.0	28.0%	5,926.2
Q3 2029	Silicon photonics and TFLN wafers	2,677.5	43.0%	1,151.3
Q3 2029	Advanced packaging, DTC and bridge services	1,912.5	38.0%	726.8
Q3 2029	12/14nm recognized service revenue	765.0	32.0%	244.8

Quarter	Revenue vector	Revenue	Gross margin	Gross profit
Q4 2029	22/28nm mainstream specialty wafers	38,955.0	40.5%	15,776.8
Q4 2029	40/65nm power, RF and connectivity wafers	32,065.0	34.5%	11,062.4
Q4 2029	90nm and older wafers	21,995.0	28.0%	6,158.6
Q4 2029	Silicon photonics and TFLN wafers	2,782.5	43.0%	1,196.5
Q4 2029	Advanced packaging, DTC and bridge services	1,987.5	38.0%	755.2
Q4 2029	12/14nm recognized service revenue	795.0	32.0%	254.4
Q1 2030	22/28nm mainstream specialty wafers	36,660.0	40.0%	14,664.0
Q1 2030	40/65nm power, RF and connectivity wafers	28,905.0	34.0%	9,827.7
Q1 2030	90nm and older wafers	18,800.0	27.5%	5,170.0
Q1 2030	Silicon photonics and TFLN wafers	3,525.0	45.0%	1,586.2
Q1 2030	Advanced packaging, DTC and bridge services	2,820.0	40.0%	1,128.0
Q1 2030	12/14nm recognized service revenue	1,175.0	34.0%	399.5
Q2 2030	22/28nm mainstream specialty wafers	38,220.0	40.0%	15,288.0
Q2 2030	40/65nm power, RF and connectivity wafers	30,135.0	34.0%	10,245.9
Q2 2030	90nm and older wafers	19,600.0	27.5%	5,390.0
Q2 2030	Silicon photonics and TFLN wafers	3,675.0	45.0%	1,653.8
Q2 2030	Advanced packaging, DTC and bridge services	2,940.0	40.0%	1,176.0
Q2 2030	12/14nm recognized service revenue	1,225.0	34.0%	416.5
Q3 2030	22/28nm mainstream specialty wafers	39,780.0	40.0%	15,912.0
Q3 2030	40/65nm power, RF and connectivity wafers	31,365.0	34.0%	10,664.1
Q3 2030	90nm and older wafers	20,400.0	27.5%	5,610.0
Q3 2030	Silicon photonics and TFLN wafers	3,825.0	45.0%	1,721.2
Q3 2030	Advanced packaging, DTC and bridge services	3,060.0	40.0%	1,224.0
Q3 2030	12/14nm recognized service revenue	1,275.0	34.0%	433.5
Q4 2030	22/28nm mainstream specialty wafers	41,340.0	40.0%	16,536.0
Q4 2030	40/65nm power, RF and connectivity wafers	32,595.0	34.0%	11,082.3
Q4 2030	90nm and older wafers	21,200.0	27.5%	5,830.0

Quarter	Revenue vector	Revenue	Gross margin	Gross profit
Q4 2030	Silicon photonics and TFLN wafers	3,975.0	45.0%	1,788.8
Q4 2030	Advanced packaging, DTC and bridge services	3,180.0	40.0%	1,272.0
Q4 2030	12/14nm recognized service revenue	1,325.0	34.0%	450.5

Operating-cost derivation

Quarter	R&D including recurring SBC	Selling and administration including SBC	Other operating income, declining grant-amortization proxy
Q3 2026	5,567.9	3,118.0	-500.0
Q4 2026	5,775.0	3,234.0	-500.0
Q1 2027	5,675.2	3,102.5	-400.0
Q2 2027	5,916.8	3,234.5	-400.0
Q3 2027	6,162.8	3,369.0	-400.0
Q4 2027	6,410.2	3,504.3	-400.0
Q1 2028	6,137.0	3,273.1	-300.0
Q2 2028	6,398.2	3,412.4	-300.0
Q3 2028	6,659.3	3,551.6	-300.0
Q4 2028	6,920.5	3,690.9	-300.0
Q1 2029	6,556.5	3,409.4	-200.0
Q2 2029	6,835.5	3,554.5	-200.0
Q3 2029	7,114.5	3,699.5	-200.0
Q4 2029	7,393.5	3,844.6	-200.0
Q1 2030	6,891.4	3,491.6	-100.0
Q2 2030	7,184.6	3,640.2	-100.0
Q3 2030	7,477.9	3,788.8	-100.0
Q4 2030	7,771.1	3,937.4	-100.0

The categories reconcile to reference operating expenses. Incremental costs below are applied once; depreciation or stock compensation already included in a reference category is not added again.

Probability-weighted financial bridge

Active = reference + sum(probability x bounded conditional delta). Negative deltas are retained. A branch with zero near-term increment remains zero only for the timing or overlap reasons documented in the evidence register.

Weighted OP below is after the stated minority operating-profit attribution. Conditional gross profit and expenses remain consolidated figures.

Quarter	Branch	p	Conditional revenue	Conditional GP	Conditional opex	Weighted OP
Q3 2026	VALUE-PRICING	70%	0.0	0.0	0.0	0.0
Q3 2026	PIC-QUALIFIED-VOLUME	60%	0.0	0.0	0.0	0.0
Q3 2026	PACKAGING-ACCEPTANCE	45%	0.0	0.0	0.0	0.0
Q3 2026	INTEL-12NM	50%	0.0	0.0	0.0	0.0
Q3 2026	MATURE-PRICE-DOWNSIDE	30%	0.0	0.0	0.0	0.0
Q4 2026	VALUE-PRICING	70%	0.0	0.0	0.0	0.0
Q4 2026	PIC-QUALIFIED-VOLUME	60%	0.0	0.0	0.0	0.0
Q4 2026	PACKAGING-ACCEPTANCE	45%	0.0	0.0	0.0	0.0
Q4 2026	INTEL-12NM	50%	0.0	0.0	0.0	0.0
Q4 2026	MATURE-PRICE-DOWNSIDE	30%	0.0	0.0	0.0	0.0
Q1 2027	VALUE-PRICING	70%	1,485.2	965.4	74.3	623.8
Q1 2027	PIC-QUALIFIED-VOLUME	60%	470.0	188.0	47.0	84.6
Q1 2027	PACKAGING-ACCEPTANCE	45%	0.0	0.0	0.0	0.0
Q1 2027	INTEL-12NM	50%	0.0	0.0	0.0	0.0
Q1 2027	MATURE-PRICE-DOWNSIDE	30%	0.0	0.0	0.0	0.0
Q2 2027	VALUE-PRICING	70%	1,548.4	1,006.5	77.4	650.3
Q2 2027	PIC-QUALIFIED-VOLUME	60%	490.0	196.0	49.0	88.2
Q2 2027	PACKAGING-ACCEPTANCE	45%	0.0	0.0	0.0	0.0
Q2 2027	INTEL-12NM	50%	0.0	0.0	0.0	0.0
Q2 2027	MATURE-PRICE-DOWNSIDE	30%	0.0	0.0	0.0	0.0
Q3 2027	VALUE-PRICING	70%	1,611.6	1,047.5	80.6	676.9

Quarter	Branch	p	Conditional revenue	Conditional GP	Conditional opex	Weighted OP
Q3 2027	PIC-QUALIFIED-VOLUME	60%	510.0	204.0	51.0	91.8
Q3 2027	PACKAGING-ACCEPTANCE	45%	0.0	0.0	0.0	0.0
Q3 2027	INTEL-12NM	50%	90.0	27.0	13.5	6.8
Q3 2027	MATURE-PRICE-DOWNSIDE	30%	0.0	0.0	0.0	0.0
Q4 2027	VALUE-PRICING	70%	1,674.8	1,088.6	83.7	703.4
Q4 2027	PIC-QUALIFIED-VOLUME	60%	530.0	212.0	53.0	95.4
Q4 2027	PACKAGING-ACCEPTANCE	45%	0.0	0.0	0.0	0.0
Q4 2027	INTEL-12NM	50%	210.0	63.0	31.5	15.8
Q4 2027	MATURE-PRICE-DOWNSIDE	30%	0.0	0.0	0.0	0.0
Q1 2028	VALUE-PRICING	70%	1,184.4	769.9	59.2	497.4
Q1 2028	PIC-QUALIFIED-VOLUME	60%	940.0	376.0	94.0	169.2
Q1 2028	PACKAGING-ACCEPTANCE	45%	0.0	0.0	0.0	0.0
Q1 2028	INTEL-12NM	50%	305.5	91.6	45.8	22.9
Q1 2028	MATURE-PRICE-DOWNSIDE	30%	-850.2	-765.2	-42.5	-216.8
Q2 2028	VALUE-PRICING	70%	1,234.8	802.6	61.7	518.6
Q2 2028	PIC-QUALIFIED-VOLUME	60%	980.0	392.0	98.0	176.4
Q2 2028	PACKAGING-ACCEPTANCE	45%	0.0	0.0	0.0	0.0
Q2 2028	INTEL-12NM	50%	318.5	95.5	47.8	23.9
Q2 2028	MATURE-PRICE-DOWNSIDE	30%	-886.4	-797.8	-44.3	-226.0
Q3 2028	VALUE-PRICING	70%	1,285.2	835.4	64.3	539.8
Q3 2028	PIC-QUALIFIED-VOLUME	60%	1,020.0	408.0	102.0	183.6
Q3 2028	PACKAGING-ACCEPTANCE	45%	400.0	140.0	48.0	41.4
Q3 2028	INTEL-12NM	50%	331.5	99.5	49.7	24.9

Quarter	Branch	p	Conditional revenue	Conditional GP	Conditional opex	Weighted OP
Q3 2028	MATURE-PRICE-DOWNSIDE	30%	-922.6	-830.3	-46.1	-235.3
Q4 2028	VALUE-PRICING	70%	1,335.6	868.1	66.8	561.0
Q4 2028	PIC-QUALIFIED-VOLUME	60%	1,060.0	424.0	106.0	190.8
Q4 2028	PACKAGING-ACCEPTANCE	45%	600.0	210.0	72.0	62.1
Q4 2028	INTEL-12NM	50%	344.5	103.3	51.7	25.8
Q4 2028	MATURE-PRICE-DOWNSIDE	30%	-958.8	-862.9	-47.9	-244.5
Q1 2029	VALUE-PRICING	70%	494.9	321.7	24.7	207.9
Q1 2029	PIC-QUALIFIED-VOLUME	60%	1,410.0	564.0	141.0	253.8
Q1 2029	PACKAGING-ACCEPTANCE	45%	705.0	246.7	84.6	73.0
Q1 2029	INTEL-12NM	50%	940.0	282.0	141.0	70.5
Q1 2029	MATURE-PRICE-DOWNSIDE	30%	-1,438.2	-1,294.4	-71.9	-366.7
Q2 2029	VALUE-PRICING	70%	516.0	335.4	25.8	216.7
Q2 2029	PIC-QUALIFIED-VOLUME	60%	1,470.0	588.0	147.0	264.6
Q2 2029	PACKAGING-ACCEPTANCE	45%	735.0	257.2	88.2	76.1
Q2 2029	INTEL-12NM	50%	980.0	294.0	147.0	73.5
Q2 2029	MATURE-PRICE-DOWNSIDE	30%	-1,499.4	-1,349.5	-75.0	-382.3
Q3 2029	VALUE-PRICING	70%	537.0	349.1	26.9	225.6
Q3 2029	PIC-QUALIFIED-VOLUME	60%	1,530.0	612.0	153.0	275.4
Q3 2029	PACKAGING-ACCEPTANCE	45%	765.0	267.8	91.8	79.2
Q3 2029	INTEL-12NM	50%	1,020.0	306.0	153.0	76.5
Q3 2029	MATURE-PRICE-DOWNSIDE	30%	-1,560.6	-1,404.5	-78.0	-398.0
Q4 2029	VALUE-PRICING	70%	558.1	362.8	27.9	234.4
Q4 2029	PIC-QUALIFIED-VOLUME	60%	1,590.0	636.0	159.0	286.2

Quarter	Branch	p	Conditional revenue	Conditional GP	Conditional opex	Weighted OP
Q4 2029	PACKAGING-ACCEPTANCE	45%	795.0	278.2	95.4	82.3
Q4 2029	INTEL-12NM	50%	1,060.0	318.0	159.0	79.5
Q4 2029	MATURE-PRICE-DOWNSIDE	30%	-1,621.8	-1,459.6	-81.1	-413.6
Q1 2030	VALUE-PRICING	70%	0.0	0.0	0.0	0.0
Q1 2030	PIC-QUALIFIED-VOLUME	60%	1,410.0	564.0	141.0	253.8
Q1 2030	PACKAGING-ACCEPTANCE	45%	1,175.0	411.2	141.0	121.6
Q1 2030	INTEL-12NM	50%	1,645.0	493.5	246.8	123.4
Q1 2030	MATURE-PRICE-DOWNSIDE	30%	-1,431.2	-1,288.0	-71.6	-364.9
Q2 2030	VALUE-PRICING	70%	0.0	0.0	0.0	0.0
Q2 2030	PIC-QUALIFIED-VOLUME	60%	1,470.0	588.0	147.0	264.6
Q2 2030	PACKAGING-ACCEPTANCE	45%	1,225.0	428.8	147.0	126.8
Q2 2030	INTEL-12NM	50%	1,715.0	514.5	257.2	128.6
Q2 2030	MATURE-PRICE-DOWNSIDE	30%	-1,492.1	-1,342.8	-74.6	-380.5
Q3 2030	VALUE-PRICING	70%	0.0	0.0	0.0	0.0
Q3 2030	PIC-QUALIFIED-VOLUME	60%	1,530.0	612.0	153.0	275.4
Q3 2030	PACKAGING-ACCEPTANCE	45%	1,275.0	446.2	153.0	132.0
Q3 2030	INTEL-12NM	50%	1,785.0	535.5	267.8	133.9
Q3 2030	MATURE-PRICE-DOWNSIDE	30%	-1,553.0	-1,397.7	-77.6	-396.0
Q4 2030	VALUE-PRICING	70%	0.0	0.0	0.0	0.0
Q4 2030	PIC-QUALIFIED-VOLUME	60%	1,590.0	636.0	159.0	286.2
Q4 2030	PACKAGING-ACCEPTANCE	45%	1,325.0	463.7	159.0	137.1
Q4 2030	INTEL-12NM	50%	1,855.0	556.5	278.2	139.1
Q4 2030	MATURE-PRICE-DOWNSIDE	30%	-1,613.9	-1,452.5	-80.7	-411.5

Consolidated profit to the common shareholder claim

Minority attribution is an earnings allocation, not operating expense or a fixed book-value deduction. Its rate is a disclosed model proxy and moves with reference profit and each probability-weighted change.

Quarter	Consolidated OP	Minority rate	Minority allocation	Owner OP
Q3 2026	17,846.3	0.0%	0.0	17,846.3
Q4 2026	18,444.0	0.0%	0.0	18,444.0
Q1 2027	19,247.6	0.0%	0.0	19,247.6
Q2 2027	20,049.6	0.0%	0.0	20,049.6
Q3 2027	20,863.4	0.0%	0.0	20,863.4
Q4 2027	21,681.1	0.0%	0.0	21,681.1
Q1 2028	20,446.5	0.0%	0.0	20,446.5
Q2 2028	21,303.8	0.0%	0.0	21,303.8
Q3 2028	22,202.5	0.0%	0.0	22,202.5
Q4 2028	23,080.5	0.0%	0.0	23,080.5
Q1 2029	21,691.1	0.0%	0.0	21,691.1
Q2 2029	22,605.6	0.0%	0.0	22,605.6
Q3 2029	23,520.1	0.0%	0.0	23,520.1
Q4 2029	24,434.6	0.0%	0.0	24,434.6
Q1 2030	22,626.3	0.0%	0.0	22,626.3
Q2 2030	23,584.9	0.0%	0.0	23,584.9
Q3 2030	24,543.4	0.0%	0.0	24,543.4
Q4 2030	25,502.0	0.0%	0.0	25,502.0

Enterprise-to-equity reconciliation

20x rolling Q+next3 TIFRS operating profit, divided by window-end Taiwan common shares including conservative new-convertible dilution. No investment-gain or excess-cash bonus.

Quarter	Reference claims	Weighted change	Total claims	Ceiling / share
Q3 2026	0.0	0.0	0.0	117.28
Q4 2026	0.0	0.0	0.0	121.92
Q1 2027	0.0	0.0	0.0	126.89
Q2 2027	0.0	0.0	0.0	128.71
Q3 2027	0.0	0.0	0.0	130.60
Q4 2027	0.0	0.0	0.0	132.63
Q1 2028	0.0	0.0	0.0	134.75
Q2 2028	0.0	0.0	0.0	136.62
Q3 2028	0.0	0.0	0.0	138.59
Q4 2028	0.0	0.0	0.0	140.57
Q1 2029	0.0	0.0	0.0	142.62
Q2 2029	0.0	0.0	0.0	144.01
Q3 2029	0.0	0.0	0.0	145.47
Q4 2029	0.0	0.0	0.0	147.00
Q1 2030	0.0	0.0	0.0	148.59
Q2 2030	0.0	0.0	0.0	-
Q3 2030	0.0	0.0	0.0	-
Q4 2030	0.0	0.0	0.0	-

Claim decisions, overlap and triggers

VALUE-PRICING

p: 0.7

hypothesis: New and repriceable differentiated wafer supply earns a residual premium after the observed 2026 improvement.

reference: Q3 2026 revenue is the actual monthly total. The reference carries that starting price/mix and a gradual 22/28 nm mix shift; it does not include a separate 2027 scarcity premium.

Conditional: 2027:4% on 50% of the three legacy wafer vectors; 2028:3% on 50%; 2029:1.5% on 40%; 2030:zero.

bounds: 2027 0-8% premium on 30-70% eligible sales. A bank analyst suggested 8-10%, but management explicitly declined a numerical confirmation.

overlap: No increment on 2026 observed revenue, fixed-price agreements, Intel revenue, photonics or packaging. Later fading prevents treating a temporary premium as permanent.

cost bridge: 65% gross-profit retention, less 5% sales/support cost =60% operating-profit conversion; the remaining 35% absorbs energy, materials, process support and customer concessions. No second utilization gain.

why p: Two management discussions confirm price-adjustment activity and stronger 2027 intentions. The amount and eligible orders are uncertain, hence 70%, not 100%.

timing: Recognized with delivered/accepted wafers under new terms during 2027; no backlog repricing assumed.

counter: Cost recovery can absorb the price rise; long-term contracts, customer resistance and Chinese capacity constrain premiums.

source ids: q1_transcript q2_transcript

PIC-QUALIFIED-VOLUME

p: 0.6

hypothesis: Qualified photonics wafers exceed the conservative reference production ramp.

reference: Reference silicon-photonics/TFLN wafer equivalents20 k/32.5 k/52.5 k/75 k in 2027/28/29/30 at NT$200000 each. They are carved out of mainstream node sales.

Conditional: Additional10 k/20 k/30 k/30 k accepted wafer equivalents at the same modeled ASP gives NT$2000 m/4000 m/6000 m/6000 m revenue.

bounds: Conditional 2028 extra0-30 k wafers; recognized wafer-equivalent ASP NT$150000-250000. These are assumptions, not quoted prices or disclosed orders.

overlap: The first SILITH delivery and the general 2027 platform are one ramp. Eight million historical SILITH PICs are not incremental UMC wafers. TFLN, pluggables and CPO share this pool.

cost bridge: 40% GP after wafer processing, depreciation and ordinary yield cost, less 10% incremental engineering/qualification/SG&A gives 30% OP.

why p: A real first production customer reduces technology risk, but broad-customer qualification, yield, capacity and ASP remain unknown.

timing: Small 2027 ramp; larger 2028-30 accepted output follows tools and qualification.

counter: Tower/GFS/TSMC alternatives, optical loss, packaging yield, customer requalification and platform readiness.

source ids: silith tfln q2_transcript

PACKAGING-ACCEPTANCE

p: 0.45

hypothesis: DTC, bridge and hybrid-bonding acceptance exceeds the reference after the new capacity becomes usable.

reference: Reference packaging revenue NT$2000 m/4500 m/7500 m/12000 m in 2027-30, including existing production and ordinary customer wins.

Conditional: Extra revenue0/1000/3000/5000 m in 2027-30; 2028 increments occur only inH2. At NT$100000 per processed-wafer service equivalent, extra 2028/29/30 volume10 k/30 k/50 k.

bounds: Extra 2028 revenue0-2000 m; unit-equivalent ASP NT$70000-130000. No complete CoWoS system revenue.

overlap: Existing DTC/bridge production remains in reference; additional revenue only after customer acceptance above it. More than35 product discussions/tape-outs are not35 revenue awards.

cost bridge: 35% GP less 12% incremental engineering and yield/acceptance support =23% OP. Full corporate fixed development remains in reference.

why p: Existing production and customer engagement are tangible, but the late 2028-29 construction/qualification window constrains success.

timing: New-facility incremental revenue beginsQ3 2028; reference existing-facility activity can grow earlier.

counter: Construction delays, customer designs migrating to competing packaging, bonding defects and customer concentration.

source ids: annual q1_transcript q2_transcript

INTEL-12NM

p: 0.5

hypothesis: Intel-based12 nm execution contributes above a small pilot/reference business.

reference: Recognized service-revenue proxy NT$200 m/1200 m/3000 m/5000 m in 2027-30. This is UMC economic revenue, not an assumed gross Intel wafer bill.

Conditional: Extra recognized revenue NT$300 m/1300 m/4000 m/7000 m in 2027-30, at NT$50000 per service-equivalent unit.

bounds: 2028 reference+conditional revenue NT$1200-3500 m; contract gross/net recognition, unit fee and profit sharing are undisclosed.

overlap: No separate14 nm, Arizona fab capacity or leading-edge TAM revenue addition. Normal12 nm development is in R&D.

cost bridge: 30% GP after partner manufacturing/economic share;15% incremental engineering and selling gives 15% OP. This is a modeling proxy pending contract economics.

why p: Process transfer and PDK progress are real, but customer tape-outs precede pilot 2027 and meaningful 2028 production; partner economics and yields remain unverified.

timing: 2027 pilot revenue is concentrated inH2;2028 ramps gradually.

counter: Tape-out delays, weak product competitiveness, partner economics, support costs and low utilization.

source ids: intel annual q1_transcript q2_transcript

MATURE-PRICE-DOWNSIDE

p: 0.3

hypothesis: China capacity and weak non-AI demand create residual price erosion beyond reference mix/fade.

reference: The older-node revenue pool is flat then declines; reference GM already fades, so this is a residual adverse case.

Conditional: 2028 price loss 3% on 60% of40/65 nm plus90 nm-and-older sales;2029-30 loss 5% on 60%; zero 2026-27.

bounds: Residual 2028 loss0-6% on40-70% of the exposed pool.

overlap: No extra unit-demand shock or utilization haircut. This overlaps part of the positive price pool but represents adverse-state residual around the reference, not mutually exclusive volume additions.

cost bridge: Lost revenue carries 90% GP loss, partly offset by 5% selling/variable cost savings:85% OP loss.

why p: The annual report identifies subsidized Chinese capacity and price pressure; differentiated processes and customer qualification mitigate, rather than eliminate, the risk.

timing: 2028-30 as capacity arrives and pricing normalizes.

counter: Persistent tight qualified capacity or faster commodity exit reduces the exposed pool.

source ids: annual

Remaining bounded uncertainties

- Q3 revenue is released monthly; Q3 margin, OP, cash and share balances are estimates until the quarterly report.

- Photonic and packaging segment sales, wafer ASP, accepted units and yields are not disclosed. Carve-outs and unit bridges are explicit assumptions.

- Intel12 nm economic allocation and gross/net revenue recognition are undisclosed. The model uses a small recognized-service revenue proxy, with a delayed 2028 meaningful ramp.

- The October5 amended filing still labels both conversion prices Tranche A. The model maps the second price to B based on the two-tranche structure and flags this unresolved textual defect.

- The October13 overseas bond issue remains future-dated as ofOctober7. No claim that proceeds are already cash.

- Full interim restricted-cash, customer-deposit and debt-maturity notes were not retrieved. Severe no-refinancing/no-new-issue stress quantifies the limit; no unconditional funded claim.

- Call reproductions contain apparent transcription errors. Original reports control financials and units; issuer webcast links are retained. The12 nm timing and pricing statements are attributed management remarks, not exact quoted contractual terms.

- No current share register later than the August10 cancellation was reconciled. Window-end shares include conservative new-CB conversion and0.15% annual award reserves.

- No3 nm acquisition/fab or standalone CPO TAM increment is modeled: counterparty, scope, timing and economic share are undefined. This is not a zero assigned merely for lack of official confirmation.

Sources and basis

The current production framework was read before initiation. Source statements are evidence, not instructions. The shared model data drives both this PDF and the linked Excel model; the clean-state editable artifact embeds these same images.

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